

Day 1 Handout: Foundations and the Potential Outcomes Framework

Experimentation and Causal Inference, SDAIA Academy. Revision notes for participants.

What Day 1 was about

Most data science answers “what is associated with what.” Decisions need the harder question: “what happens if we intervene?” Day 1 gave you the framework that makes “the effect” a precise object, the assumptions under which it can be recovered from data, and the reason randomization is the benchmark against which every other method is judged.

Three kinds of questions

Association (seeing): $P(Y \text{ given } X)$. What does observing X tell me about Y ? Prediction and most machine learning live here.

Intervention (doing): $P(Y \text{ given } \text{do}(X))$. What happens to Y if I set X ? The language of experiments and policy.

Counterfactual (imagining): what would this unit’s Y have been had X been different? The basis of individual effects.

Conditioning filters the world as it is; intervening changes it. Confusing the two is the root of most causal errors in analytics. A better predictive model does not climb the ladder.

Why correlation is not causation

An association between treatment and outcome can come from four stories, and only one is an effect: reverse causation (Y causes X), confounding (a common cause drives both), selection (conditioning on how units entered the sample, also called collider bias), and chance. Chance is the only one that more data fixes. Bias is a property of design and identification, not of sample size: a confounded comparison converges to the wrong number as n grows.

Injaz example. Users who opted into the new guided uploader complete services at a higher rate. They are also younger, more digitally literate and on newer devices, which is exactly the profile that completes more anyway. The comparison measures who chose the feature, not what the feature did.

The workflow (the spine of the whole course)

1. Question to estimand: translate the decision into a precise causal quantity (whose effect, of what, on whom, on which outcome).
2. Identification: state the assumptions under which that estimand equals something computable from observed data.
3. Estimation: choose an estimator and fit it. It is only as good as the identification behind it.
4. Inference: quantify uncertainty with the right standard error for the design.
5. Sensitivity: ask how far the assumptions could be wrong before the conclusion flips, and report it.

Vocabulary to use precisely: unit, treatment T , outcome Y , covariates X (pre-treatment only; never post-treatment variables in the adjustment set), estimand (the true quantity), estimator (the procedure), estimate (the number).

Potential outcomes (Neyman-Rubin)

Each unit i carries two potential outcomes: $Y_{i(1)}$ if treated and $Y_{i(0)}$ if not. Both exist in principle; we observe one.

Individual effect: $\tau_i = Y_{i(1)} - Y_{i(0)}$. A within-unit contrast, not a between-group difference.

Observed outcome: $Y_i = T_i Y_{i(1)} + (1 - T_i) Y_{i(0)}$. Treatment selects which potential outcome we see.

The fundamental problem: we never observe both potential outcomes for the same unit, so τ_i is unobservable.

Causal inference is a missing-data problem. Averages rescue us: the average effect can be identified under the right design even though no individual effect can.

Estimands:

Name	Definition	When it is the right target
ATE	$E[Y(1) \text{ minus } Y(0)]$ over everyone	“Launch to everyone versus no one”
ATT	$E[Y(1) \text{ minus } Y(0) \text{ given } T = 1]$	Evaluating a program delivered to a specific or self-selected group
CATE	$E[Y(1) \text{ minus } Y(0) \text{ given } X = x]$	Targeting, personalization, fairness
ITT	Effect of being assigned, regardless of uptake	What a real rollout can deliver

Name the estimand before touching data. “The effect” is ambiguous; “the ATT on completion rate for training attendees” is answerable.

Selection bias, made precise

$$E[Y \text{ given } T = 1] \text{ minus } E[Y \text{ given } T = 0] = \text{ATT} + \{ E[Y(0) \text{ given } T = 1] \text{ minus } E[Y(0) \text{ given } T = 0] \}$$

Read it aloud: the observed gap equals the real effect on the treated plus how different the groups would have been even without treatment. The braces are selection bias. If treatment is independent of the potential outcomes, that term is zero and the naive difference equals the ATE. Design is the fight to zero out the second term.

Worked example from the deck. Six users. $Y(1) = [30, 25, 40, 22, 35, 28]$, $Y(0) = [24, 26, 31, 20, 27, 25]$. True ATE = mean of the differences = +4.17. If units 1, 3, 5 are treated we observe treated mean 35 and control mean 23.7, a naive +11.3, inflated 2.7 times because the treated units were healthier to begin with.

Common misreadings. $Y(1)$ and $Y(0)$ are defined for every unit, including units never treated. ATE = 0 does not mean no effect for anyone; positive and negative CATEs can cancel. Potential outcomes are fixed attributes; randomness enters through which one we observe.

The identification assumptions

	Assumption	Meaning	Testable?
A1	SUTVA	No interference between units; one well-defined version of treatment	Partly, from domain knowledge
A2	Ignorability (unconfoundedness)	$\{Y(1), Y(0)\}$ independent of T given X	No, in observational data. Holds by design in an experiment
A3	Positivity (overlap)	$0 < P(T = 1 \text{ given } X = x) < 1$ for every x	Yes, plot propensity scores by arm
A4	Consistency	Observed Y equals $Y(T)$	Assumed

SUTVA breaks with network effects, marketplaces, shared devices and hidden treatment variants. Positivity breaks structurally (a rule forbids treatment for a group) or practically (propensities near 0 or 1). Ignorability is the load-bearing wall: the data cannot reveal a confounder you never measured, which is why every observational estimate needs a sensitivity analysis.

Good controls, bad controls. Confounder (common cause of T and Y): adjust. Mediator (on the path T to M to Y): do not adjust for a total effect. Collider (common effect): do not adjust; conditioning opens a spurious path. Anything post-treatment: never adjust.

Under A1 to A4 the ATE becomes computable: $\text{ATE} = E_X[E[Y \text{ given } T = 1, X] \text{ minus } E[Y \text{ given } T = 0, X]]$ (adjustment, the g-formula), or by re-weighting units by $1/P(T \text{ given } X)$ (IPW). Doubly robust estimators combine both and stay consistent if either model is right (Day 3).

Why randomization works

Random assignment makes ignorability true by construction: assignment is independent of everything, so $\{Y(1), Y(0)\}$ is independent of T with no covariates required. It balances measured and unmeasured confounders in expectation, including the ones you never thought of. The control group becomes an unbiased stand-in for the treated group's missing $Y(0)$.

Balance holds in expectation, not in every draw. A single experiment can show chance imbalance, especially at small n . Never drop units to force balance after the fact. Stratify or block on strong predictors before randomizing, and adjust for pre-treatment covariates in analysis.

Two inference philosophies. Design-based (Fisher, Neyman): potential outcomes are fixed, only T is random, no outcome model. Model-based: units are a sample from a population and inference goes through an assumed model. Design-based is the assumption-light default for experiments.

Fisher's sharp null: $Y_i(1) = Y_i(0)$ for every unit. Under it both potential outcomes are known for everyone, so the test statistic can be recomputed under every re-assignment. The p-value is the fraction of re-randomizations at least as extreme as observed. Exact, no normality, no large n .

Neyman: the difference in arm means is unbiased for the ATE under complete randomization; its variance S_1^2 squared over n_1 plus S_0^2 squared over n_0 is conservative. In practice, report the difference in means with a robust (HC2) standard error.

When you cannot randomize (ethics, feasibility, cost and time, history), quasi-experiments borrow the logic of randomization from natural or as-if-random variation. That is Days 3 and 4.

Lab 1 recap

You built a population with both potential outcomes, planted a 5-point effect, and saw the naive self-selected estimate land near 9 to 10 points while the randomized estimate landed near 5. You computed the selection-bias term directly, plotted covariate balance under both assignments, ran a randomization test with 1,000 permutations, and swept n to one million: the biased estimate converged confidently to the wrong number. More data reduces variance, not bias.

Self-check (answers at the end)

1. State the fundamental problem of causal inference in one sentence.
2. Decompose the naive difference in means.
3. A regulator asks how much emissions fell among firms that adopted a standard, because of the standard. Which estimand?
4. A viral referral program means treated users recruit control users. Which assumption is threatened?
5. No one over 65 was ever eligible for treatment. Which assumption is violated?
6. True or false: with enough data, selection bias disappears.
7. What does Fisher's randomization test rely on for validity?
8. Why should the adjustment set contain only pre-treatment variables?

Answers

1. For any unit only one potential outcome is observed, so individual effects are unobservable.
2. ATT plus selection bias, where bias = $E[Y(0) \text{ given } T = 1] - E[Y(0) \text{ given } T = 0]$.
3. ATT.
4. SUTVA (interference).
5. Positivity (structural violation).
6. False. Large n shrinks variance, not bias.
7. The assignment mechanism itself, not normality or the CLT.
8. Post-treatment variables can be mediators or colliders; adjusting for them injects bias.